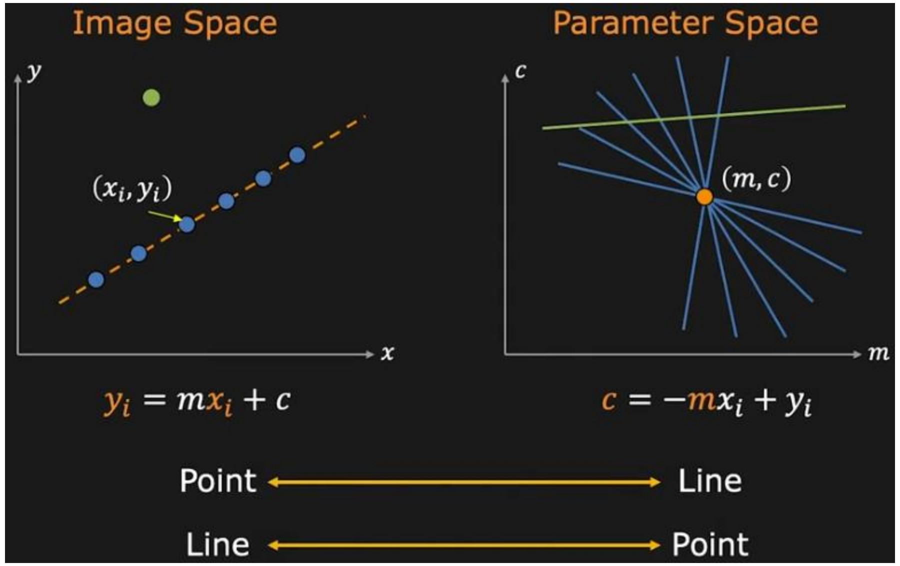
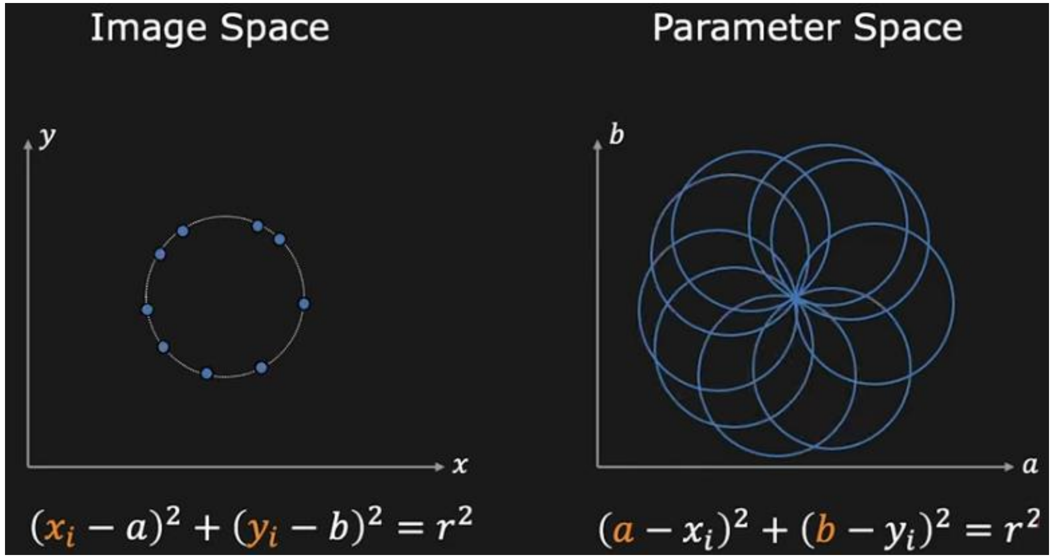
Hough Transform

- We know how to detect:

Straight lines



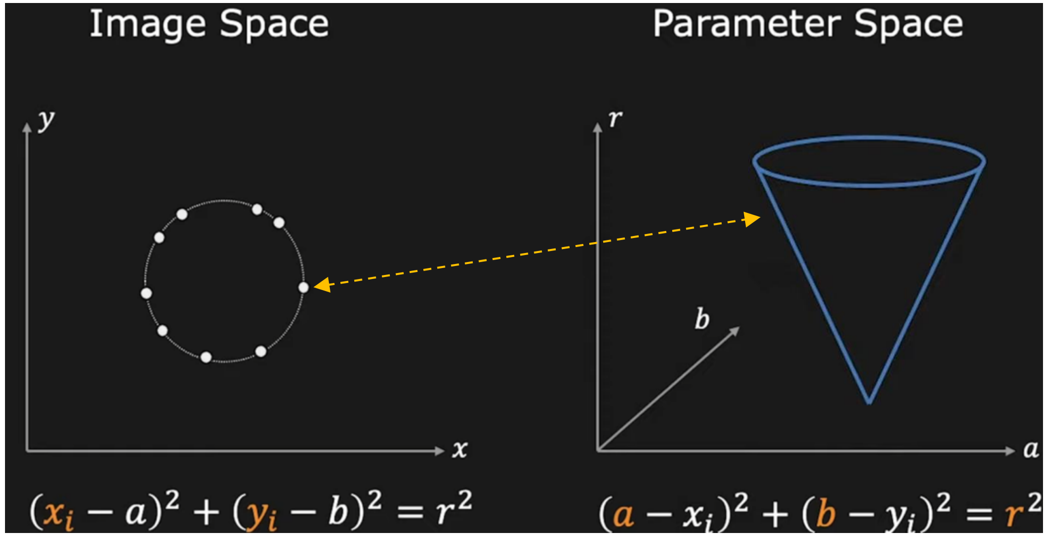
Circles (r known)



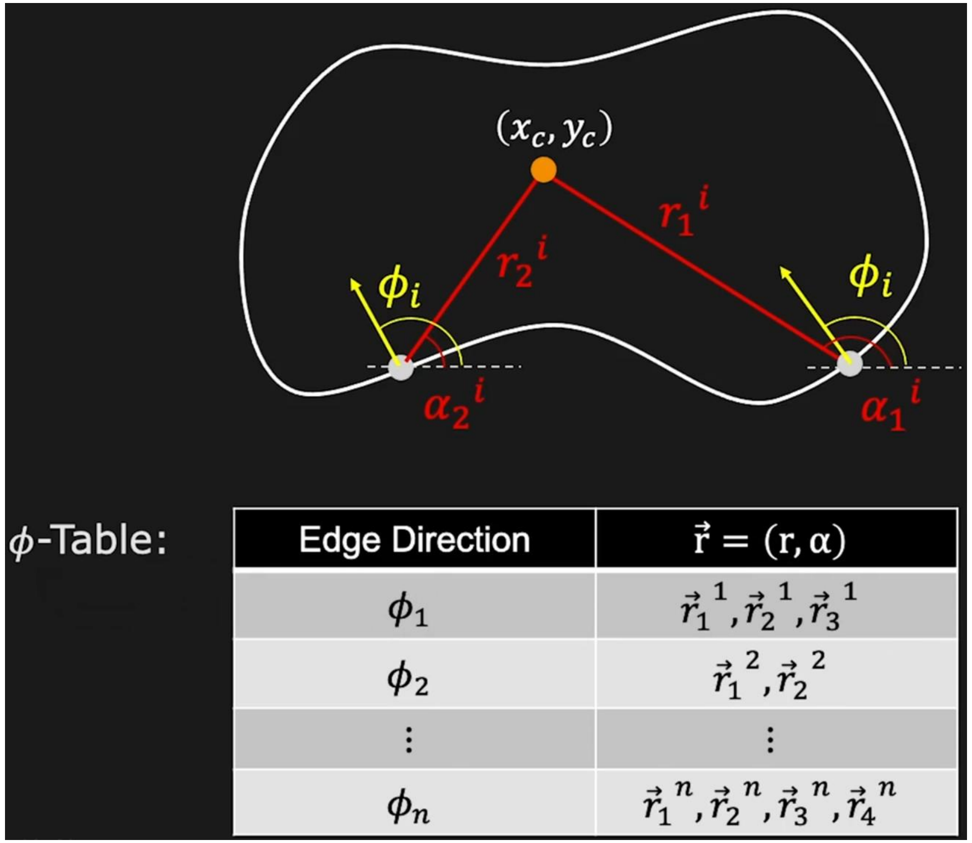
Hough Transform

- We know how to detect:

Circles (r unknown)

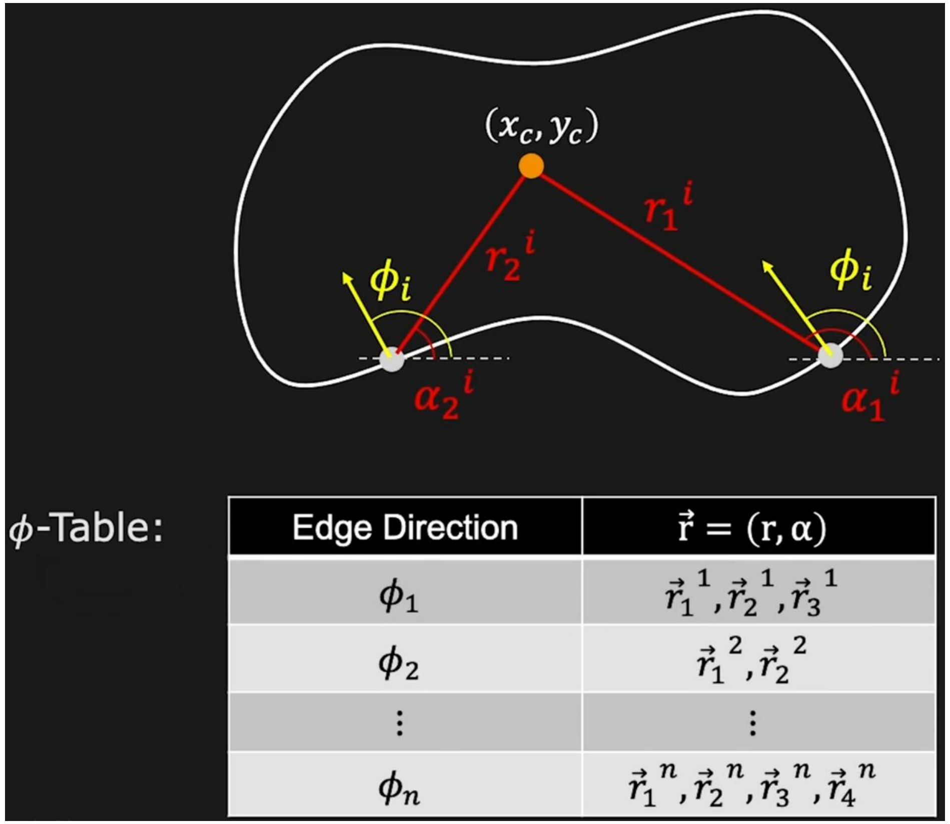


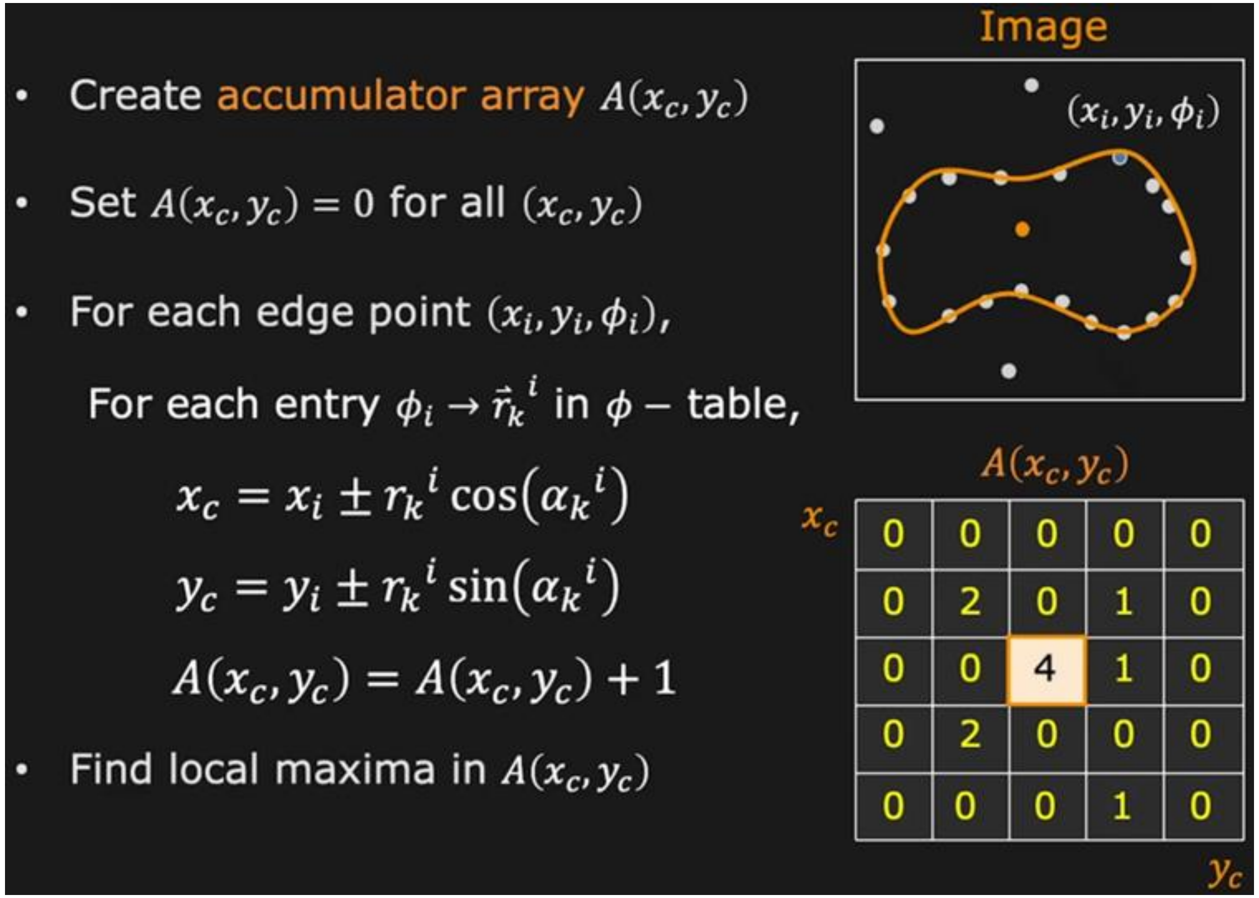
Arbitrary shapes
(parameterized following the Generalized Hough Transform)



Hough Transform

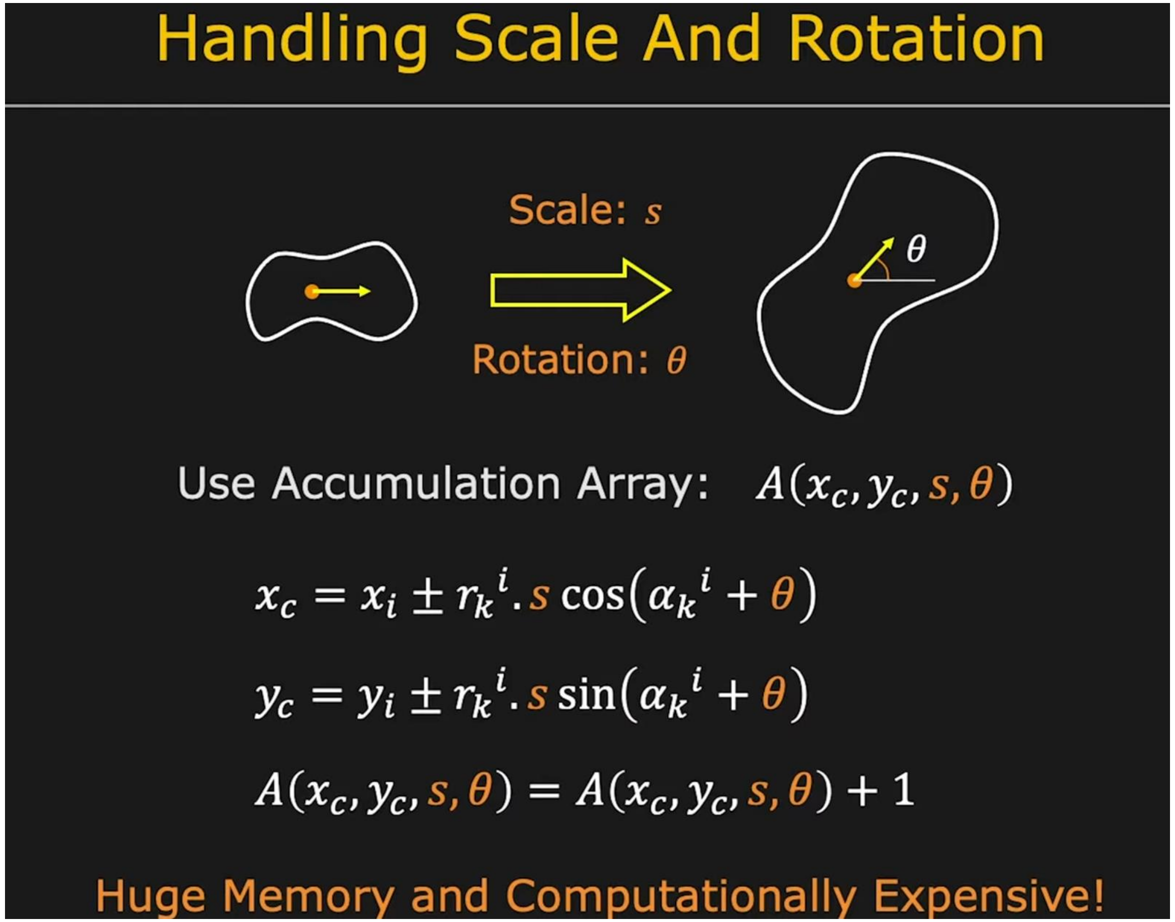
- In the context of the Generalized Hough Transform,
  - what happens if we want to deal as well with rotation and scale?? **We know that our accumulator increases dimensionality, what should we also change our φ-Table?**





Hough Transform

- In the context of the generalized Hough Transform,
  – **We know that our accumulator increases dimensionality, what should we also change our φ-Table?**



The φ-Table doesn't change!

The space of possible scales and rotations is sampled to compute x_c and y_c.

Generalized Hough Transform | Boundary Detection (Shree Nayar)

Hough Transform

- In the context of the generalized Hough Transform,
  - **We know that our accumulator increases dimensionality, what should we also change our φ-Table?**

```
for each edge point (xi, yi, φi) in the image:
    for each scale s:
        for each rotation θ:
            φ_model = φi - θ
            for each entry (rk, αk) in R_table[φ_model]:
                xc = xi + rk * s * cos(αk + θ)
                yc = yi + rk * s * sin(αk + θ)
                if (xc, yc) inside bounds:
                    A[xc, yc, s, θ] += 1
```

The φ-Table doesn't change!

The space of possible scales and rotations is sampled to compute x_c and y_c.

Generalized Hough Transform | Boundary Detection (Shree Nayar)